

PROJECT REPORT

ON

CYBER CRIME ANALYSIS



Submitted to

LKCTC

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CERTIFICATE

This is to certify that the project report entitled "Cyber Crime Analysis" submitted by Lovepreet Kaur in partial fulfillment of the requirements for the degree of Bachelor of Technology (Artificial Intelligence and Machine Learning) at Lyallpur Khalsa College Technical Campus, Jalandhar, is a bonafide record of the work carried out under the guidance of Mr. Harkirat Singh (Data Scientist).

To the best of my knowledge, this project has not been submitted, either in part or in full, to any other university or institution for the award of any degree or diploma.

Date: _____

Place: Jalandhar

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I would also like to thank my parents, friends, and classmates for their constant support and encouragement throughout the completion of this project.

Finally, I express my gratitude to everyone who directly or indirectly contributed to the successful completion of this project.

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ABSTRACT

Cyber crime has become one of the fastest-growing challenges in the digital world. With the increasing use of the internet, online banking, e-commerce, social media, and cloud services, cyber attacks have also increased significantly. The objective of this project is to analyze cyber crime data and identify important trends, patterns, and risk factors using data analytics and visualization techniques.

This project uses a cyber crime dataset containing information such as Country, Year, Attack Source, Target Industry, Financial Loss, Number of Affected Users, Security Vulnerability Type, Defense Mechanism Used, and Incident Resolution Time. The data is cleaned, processed, and analyzed using Python libraries such as Pandas and Plotly, while Streamlit is used to develop an interactive dashboard.

Various visualizations such as bar charts, pie charts, treemaps, line charts, and KPI cards are used to present meaningful insights. The dashboard helps users understand cyber crime trends, identify high-risk industries, compare financial losses, and evaluate the effectiveness of different defense mechanisms.

The project demonstrates how data analytics can support organizations in improving cyber security awareness, making informed decisions, and reducing the impact of cyber attacks. It also highlights the importance of visualization in simplifying complex cyber security data.

Keywords: Cyber Crime, Data Analysis, Python, Streamlit, Data Visualization, Cyber Security, Dashboard.

TABLE OF CONTENTS

Chapter	
Title	
Page No.	
i.Certificate	
ii.Acknowledgement	
iii.Abstract	
iv.Chapter 1	
Introduction	
1.1 Introduction to Cyber Crime	
1.2 Problem Statement	
1.3 Objectives of the Project	
1.4Scope of the Project	
V.Chapter 2	
Literature Review	
V1.Chapter 3	
Methodology	
3.1Data Collection	
3.2 Data Preprocessing	
3.3 Exploratory Data Analysis	
3.4 Dashboard Development	
Vii.Chapter 4	
Implementation	
Viii. Chapter 5	
Results and Analysis	
Chapter 6	
Conclusion	
Chapter 7	
Future Scope	

CHAPTER 1

INTRODUCTION

Cyber crime refers to illegal activities carried out using computers, mobile devices, networks, or the internet. With the rapid growth of digital technology, people increasingly rely on online services such as internet banking, e-commerce, social media, cloud computing, and digital payments. While these technologies provide convenience and efficiency, they also create opportunities for cyber criminals to exploit vulnerabilities.

Cyber attacks can result in financial loss, data theft, identity theft, privacy violations, and disruption of business operations. Individuals, businesses, and government organizations are all potential targets of cyber crime. Therefore, understanding cyber crime patterns and analyzing attack trends has become essential for improving cyber security and protecting digital assets.

The Cyber Crime Analysis project focuses on analyzing cyber crime data to identify important trends, affected industries, attack sources, financial losses, and commonly used defense mechanisms. The project uses data analytics and visualization techniques to transform raw data into meaningful insights. Python libraries such as Pandas and Plotly are used for data analysis and visualization, while Streamlit is used to develop an interactive dashboard.

This project helps users understand cyber crime more effectively by presenting information through charts, graphs, and KPI cards. The insights generated from the analysis can support better decision-making and contribute to stronger cyber security practices.

1.1 Introduction to Cyber Crime

Cyber crime is any unlawful activity that involves the use of computers, digital devices, networks, or the internet. It includes crimes in which computers are used as a tool to commit illegal activities or are themselves the target of an attack. As digital technologies continue to grow, cyber crime has become one of the most significant challenges for individuals, businesses, and governments around the world.

Common types of cyber crime include phishing, malware attacks, ransomware, hacking, identity theft, data breaches, online financial fraud, denial-of-service (DoS) attacks, and cyber stalking. These attacks can cause financial losses, damage an organization's reputation, expose confidential information, and disrupt essential services.

To reduce cyber threats, organizations implement various cyber security measures such as firewalls, antivirus software, Virtual Private Networks (VPNs), encryption, multi-factor authentication (MFA), intrusion detection systems, and regular security updates. However, as cyber criminals continuously develop new attack techniques, cyber security must also evolve to address emerging threats.

In this project, cyber crime data is analyzed using Python and visualized through an interactive Streamlit dashboard. The analysis helps identify trends in cyber attacks, affected industries, attack sources, financial losses, and defense mechanisms. These insights can support better decision-making and improve cyber security awareness.

1.1 Problem Statement

The rapid growth of internet usage, online banking, digital payments, e-commerce, and cloud computing has significantly increased the number of cyber attacks worldwide. Every day, organizations and individuals face threats such as phishing, ransomware, malware, data breaches, identity theft, and financial fraud. These incidents lead to financial losses, compromise sensitive information, and disrupt business operations.

Large volumes of cyber crime data are generated every year, making manual analysis difficult and time-consuming. Without proper analysis and visualization, it becomes challenging to identify attack patterns, affected industries, high-risk regions, common vulnerabilities, and the effectiveness of different security measures.

The main problem addressed in this project is the lack of a simple and interactive system for analyzing cyber crime data. To solve this problem, a data analytics dashboard has been developed using Python and Streamlit. The dashboard presents cyber crime information through interactive charts, graphs, and KPI cards, enabling users to identify trends, compare incidents, and make informed decisions for improving cyber security.

This project helps transform raw cyber crime data into meaningful insights that can support organizations in strengthening their security strategies and reducing the impact of cyber threats.

1.1 Objectives of the Project

The primary objective of this project is to analyze cyber crime data and present meaningful insights through data visualization and an interactive dashboard. The project aims to help users understand cyber crime trends, identify risk factors, and support better decision-making in the field of cyber security.

The specific objectives of this project are:

To collect and analyze cyber crime data from a reliable dataset.

To clean and preprocess the data for accurate analysis.

To identify trends and patterns in cyber crime incidents across different years and countries.

To analyze attack sources, target industries, financial losses, and the number of affected users.

To evaluate the effectiveness of different defense mechanisms used against cyber attacks.

To create interactive visualizations using Plotly for better understanding of the data.

To develop a user-friendly dashboard using Streamlit that allows users to explore cyber crime data interactively.

To provide useful insights that can help organizations improve cyber security awareness and decision-making.

The successful completion of this project demonstrates how data analytics and visualization techniques can simplify complex cyber crime data and support effective cyber security planning.

1.2 Scope of the Project

The scope of this project is to analyze cyber crime data and present meaningful insights through an interactive dashboard. The project focuses on understanding cyber crime trends by examining different factors such as attack sources, target industries, financial losses, affected users, security vulnerabilities, and defense mechanisms. The analysis helps users identify patterns and make data-driven decisions related to cyber security.

The dashboard enables users to explore cyber crime data using various visualizations, including bar charts, pie charts, treemaps, line charts, and KPI cards. These visualizations simplify complex datasets and make the information easier to understand. The project can be useful for students, researchers, organizations, and cyber security professionals who want to study cyber crime trends and evaluate the impact of different cyber attacks.

Although the project is based on a specific dataset, the same approach can be applied to larger and real-time cyber security datasets in the future. Additional features such as machine learning models for cyber attack prediction, real-time data integration, automated threat detection, and advanced analytics can further enhance the system.

This project provides a strong foundation for understanding cyber crime through data analysis and demonstrates the importance of visualization in improving cyber security awareness and decision-making.

CHAPTER 2

LITERATURE REVIEW

Cyber crime has become one of the most important areas of research due to the rapid growth of digital technologies and internet-based services. Researchers have studied various types of cyber attacks, including phishing, malware, ransomware, identity theft, and data breaches, to understand their causes, impact, and prevention techniques. Several studies have emphasized that cyber attacks not only cause financial losses but also affect the privacy, reputation, and operational efficiency of organizations.

Many researchers have used data analytics, machine learning, and visualization techniques to analyze cyber crime datasets. These methods help identify attack patterns, high-risk sectors, common vulnerabilities, and trends over different years. Interactive dashboards have also been widely adopted because they allow users to explore data easily and support faster decision-making.

Recent studies suggest that visualization tools play an important role in simplifying complex cyber security data. Charts, graphs, and KPI indicators enable users to understand cyber threats more effectively than traditional reports. Technologies such as Python, Pandas, Plotly, and Streamlit are commonly used to build interactive cyber crime analysis systems.

This project builds upon these concepts by developing a user-friendly dashboard that analyzes cyber crime data and presents meaningful insights. The dashboard allows users to examine cyber crime trends, compare attack sources, evaluate financial losses, and understand the effectiveness of different defense mechanisms.

2.1 Existing System

The existing system for cyber crime analysis mainly relies on static reports, spreadsheets, and manual data analysis. Although these methods provide basic information, they are often time-consuming and do not offer interactive features for exploring large datasets. As the volume of cyber crime incidents increases every year, traditional methods become less effective in identifying hidden trends and patterns.

Most existing systems present data in the form of tables or simple charts, making it difficult for users to compare different parameters such as attack sources, target industries, financial losses, and defense mechanisms. They also lack real-time filtering and interactive visualization, which limits the user's ability to perform detailed analysis.

Another limitation of traditional systems is that they require significant manual effort to prepare reports and summarize data. This process increases the chances of errors and makes decision-making slower.

To overcome these limitations, this project introduces an interactive dashboard developed using Python and Streamlit. The dashboard provides dynamic visualizations, KPI cards, and filtering options that enable users to analyze cyber crime data efficiently. It helps users gain meaningful insights quickly and supports better understanding of cyber crime trends.

2.2 Proposed System

The proposed system is an interactive Cyber Crime Analysis Dashboard developed using Python and Streamlit. It is designed to simplify the analysis of cyber crime data by transforming raw information into meaningful visual insights. Unlike traditional reporting methods, the proposed system provides an easy-to-use interface that allows users to explore cyber crime data efficiently.

The system begins by collecting the cyber crime dataset and performing data preprocessing, including handling missing values, removing duplicates, and formatting the data for analysis. After preprocessing, the data is analyzed using Pandas, while Plotly is used to create interactive charts and visualizations. The final dashboard is developed using Streamlit, enabling users to interact with the data through filters and graphical reports.

The dashboard displays important information such as attack sources, target industries, financial losses, affected users, security vulnerabilities, defense mechanisms, and incident resolution time. It also includes KPI cards and multiple chart types, including bar charts, pie charts, treemaps, line charts, and scatter plots, to help users understand cyber crime trends more effectively.

The proposed system provides a faster, more accurate, and user-friendly approach to cyber crime analysis. It supports better decision-making by presenting complex data in a clear and interactive format, making it valuable for students, researchers, organizations, and cyber security professionals.

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2.3 Advantages of the Proposed System

The proposed Cyber Crime Analysis Dashboard offers several advantages over traditional methods of cyber crime analysis. It provides an interactive and user-friendly platform that simplifies the understanding of complex cyber crime data. By using modern data analytics and visualization techniques, the system enables users to explore information more efficiently and accurately.

One of the major advantages of the system is its ability to present large amounts of data through interactive charts, graphs, and KPI cards. Users can quickly identify cyber crime trends, compare attack sources, analyze financial losses, and study the impact on different industries. The filtering feature allows users to focus on specific countries, years, or attack categories, making the analysis more flexible.

The system also reduces the time and effort required for manual data analysis. Since the dashboard automatically processes and visualizes the dataset, users can obtain meaningful insights without performing complex calculations or preparing lengthy reports. This improves both efficiency and accuracy.

Furthermore, the dashboard supports better decision-making by helping organizations understand cyber security risks and evaluate the effectiveness of different defense mechanisms. It can also be used for educational purposes, research, and awareness programs related to cyber security.

Overall, the proposed system provides a reliable, efficient, and scalable solution for cyber crime analysis, making it more effective than traditional reporting methods.

CHAPTER 3

METHODOLOGY

The methodology describes the systematic process followed to complete the Cyber Crime Analysis project. It includes data collection, data preprocessing, exploratory data analysis (EDA), data visualization, dashboard development, and result interpretation. A structured approach ensures that the analysis is accurate, reliable, and easy to understand.

The first step of the project is the collection of the cyber crime dataset from a reliable source. The dataset contains information such as Country, Year, Attack ID, Target Industry, Financial Loss (USD), Number of Affected Users, Attack Source, Security Vulnerability Type, Defense Mechanism Used, and Incident Resolution Time (Hours).

After collecting the dataset, data preprocessing is performed to improve data quality. This includes handling missing values, removing duplicate records, correcting data types, and preparing the dataset for analysis. Clean and organized data helps produce accurate results and meaningful visualizations.

Exploratory Data Analysis (EDA) is then carried out using Python and the Pandas library to identify trends, relationships, and patterns in the dataset. Various interactive visualizations are created using Plotly, and the final dashboard is developed using Streamlit. The dashboard allows users to interact with the data, apply filters, and analyze cyber crime trends effectively. The methodology adopted in this project provides a systematic and efficient approach for converting raw cyber crime data into meaningful insights, thereby supporting better understanding and informed decision-making.

3.1 Data Collection

Data collection is the first and one of the most important steps in the Cyber Crime Analysis project. The quality of the analysis depends on the accuracy and completeness of the collected data. For this project, a cyber crime dataset containing records from 2018 to 2024 was used to study cyber attack trends and patterns.

The dataset includes several important attributes such as Country, Year, Attack ID, Target Industry, Financial Loss (USD), Number of Affected Users, Attack Source, Security Vulnerability Type, Defense Mechanism Used, and Incident Resolution Time (Hours). These attributes provide detailed information about cyber crime incidents and help in understanding different aspects of cyber attacks.

The collected dataset was stored in CSV (Comma-Separated Values) format, which is easy to process using Python. The CSV file was imported into the Python environment using the Pandas library for further analysis and visualization.

The collected data serves as the foundation of the project and enables the identification of cyber crime trends, affected industries, financial losses, common attack sources, and the effectiveness of different security measures. Proper data collection ensures reliable analysis and supports the development of an accurate and interactive dashboard.

3.2 Data Preprocessing

Data preprocessing is an essential step in data analysis because raw data often contains missing values, duplicate records, inconsistent formats, and incorrect data types. These issues can affect the accuracy of the analysis and lead to misleading results. Therefore, the dataset was cleaned and prepared before performing any analysis.

The preprocessing process began by importing the dataset into Python using the Pandas library. The dataset was carefully examined to identify missing values and duplicate records. Missing values were handled appropriately, while duplicate entries were removed to improve data quality. The data types of different columns were also verified and corrected wherever necessary.

After cleaning the dataset, the column names were standardized and unnecessary spaces or formatting issues were removed. The dataset was then checked for consistency to ensure that all records were accurate and suitable for analysis. This step improved the reliability of the dataset and reduced the possibility of errors during visualization.

Finally, the cleaned dataset was prepared for Exploratory Data Analysis (EDA) and dashboard development. Proper data preprocessing ensured that the visualizations and insights generated by the project were accurate, meaningful, and reliable, providing a strong foundation for effective cyber crime analysis.

3.3 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is the process of examining and understanding a dataset before applying advanced analysis techniques. It helps identify patterns, trends, relationships, and anomalies within the data. In this project, EDA was performed to gain meaningful insights from the cyber crime dataset and to prepare the data for visualization.

The dataset was analyzed using the Pandas library to calculate summary statistics, examine different variables, and identify trends in cyber crime incidents. Various attributes such as Country, Year, Attack Source, Target Industry, Financial Loss (USD), Number of Affected Users, Security Vulnerability Type, Defense Mechanism Used, and Incident Resolution Time (Hours) were studied to understand their distribution and relationships.

Several interactive visualizations were created using the Plotly library. These included bar charts, pie charts, line charts, treemaps, scatter plots, and KPI cards. These visualizations helped compare cyber attacks across different countries and years, identify the most affected industries, analyze financial losses, evaluate defense mechanisms, and observe trends in incident resolution time.

The insights obtained from EDA formed the basis of the interactive Streamlit dashboard. The analysis made complex cyber crime data easier to understand and supported effective decision-making by presenting information in a clear and visually appealing manner.

3.4 Dashboard Development

The dashboard was developed using Streamlit, an open-source Python framework for building interactive web applications. Streamlit provides a simple and efficient way to convert data analysis into a user-friendly dashboard without requiring advanced web development skills.

The dashboard was designed to present cyber crime data in an interactive and visually appealing format.

The cleaned dataset was imported into the Streamlit application using the Pandas library.

Various interactive charts were created using Plotly to display important information related to cyber crime incidents. A sidebar was added to allow users to filter data based on different parameters such as Country, Year, Target Industry, Attack Source, and Defense Mechanism. These filters help users analyze specific data according to their requirements.

To provide a quick overview of the dataset, KPI (Key Performance Indicator) cards were included to display important statistics such as the total number of incidents, total financial loss, total affected users, and the average incident resolution time. Different visualizations, including bar charts, line charts, pie charts, treemaps, scatter plots, and sunburst charts, were used to present cyber crime trends and comparisons effectively.

The final Streamlit dashboard provides an interactive platform for exploring cyber crime data. It enables users to understand complex information easily, identify important trends, and make informed decisions related to cyber security. The dashboard is efficient, responsive, and easy to use, making it suitable for students, researchers, and organizations.

IMPLEMENTATION

The implementation phase focuses on developing the Cyber Crime Analysis Dashboard using Python and its supporting libraries. The project was implemented in a systematic manner, starting from data collection and preprocessing to visualization and dashboard development. Each step was carefully executed to ensure accurate analysis and meaningful results.

The project was developed using Python as the programming language. The cyber crime dataset, stored in CSV format, was imported into the Python environment using the Pandas library. Data preprocessing techniques such as handling missing values, removing duplicate records, and correcting data types were applied to improve data quality before analysis.

After preprocessing, Plotly was used to create interactive visualizations, including bar charts, line charts, pie charts, treemaps, scatter plots, sunburst charts, and KPI cards. These visualizations help users understand cyber crime trends, compare financial losses, analyze attack sources, and identify the most affected industries.

The final dashboard was developed using Streamlit, which provides an interactive interface for users to explore the dataset. The dashboard includes a sidebar with filters, KPI cards for summary statistics, and multiple interactive charts. Users can filter the data based on different parameters and instantly view updated visualizations, making the analysis more effective and user-friendly.

The implementation of this project successfully demonstrates how Python-based data analytics and visualization tools can transform raw cyber crime data into meaningful insights that support cyber security awareness and decision-making.

4.1 Software Requirements

The Cyber Crime Analysis project was developed using Python and modern data visualization tools. The software requirements were selected to ensure efficient data processing, visualization, and dashboard development.

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The project was developed using Python as the programming language. Pandas was used for data collection, cleaning, and preprocessing, while Plotly was used to create interactive charts and graphs. The interactive dashboard was developed using Streamlit, allowing users to explore cyber crime data through filters and visualizations. The project was implemented in Visual Studio Code (VS Code), and the dataset was stored in CSV format.

Software Used:

Programming Language: Python

IDE: Visual Studio Code (VS Code)

Framework: Streamlit

Libraries: Pandas, Plotly

Dataset Format: CSV

Operating System: Windows

These software tools provided an efficient environment for analyzing cyber crime data and developing an interactive dashboard.

4.2 System Implementation

The Cyber Crime Analysis Dashboard was implemented using a step-by-step approach. First, the cyber crime dataset was imported into Python using the Pandas library. The data was cleaned by removing duplicate records, handling missing values, and correcting data types to improve data quality.

After preprocessing, the data was analyzed and visualized using the Plotly library. Various charts such as bar charts, pie charts, line charts, treemaps, scatter plots, and KPI cards were created to represent cyber crime trends and statistics in an easy-to-understand format.

Finally, the dashboard was developed using Streamlit. Interactive filters were added to the sidebar, allowing users to analyze the data based on different parameters such as Country, Year, Target Industry, Attack Source, and Defense Mechanism. The dashboard provides meaningful insights and enables users to explore cyber crime data efficiently.

The implementation successfully transformed raw data into an interactive dashboard that supports better understanding and analysis of cyber crime trends.

CHAPTER 5

RESULTS AND ANALYSIS

The results of the Cyber Crime Analysis project are presented through an interactive Streamlit dashboard. The dashboard provides meaningful insights into cyber crime trends using various visualizations and summary statistics. The analysis helps users understand attack patterns, affected industries, financial losses, and the effectiveness of different defense mechanisms.

The dashboard includes KPI cards that display important information such as the total number of incidents, total financial loss, total affected users, and average incident resolution time. Interactive filters allow users to analyze the data based on different countries, years, attack sources, and target industries.

Various charts such as bar charts, line charts, pie charts, treemaps, scatter plots, and sunburst charts are used to present the analysis clearly. These visualizations make it easier to identify cyber crime trends, compare different categories, and support informed decision-making.

Overall, the dashboard successfully converts raw cyber crime data into meaningful visual insights, making cyber crime analysis simple, interactive, and effective.

5.1 Key Performance Indicators (KPI Cards)

The dashboard begins with Key Performance Indicator (KPI) cards, which provide a quick summary of the cyber crime dataset. These cards display important information such as the total number of cyber crime incidents, total financial loss, total number of affected users, and average incident resolution time. KPI cards help users understand the overall status of the dataset without analyzing each record individually.

The KPI section improves the readability of the dashboard by presenting important statistics in a simple and organized format. It enables users to identify the overall impact of cyber crime and supports faster decision-making. The values displayed in the KPI cards are automatically updated whenever the user applies filters such as Country, Year, Attack Source, or Target Industry.

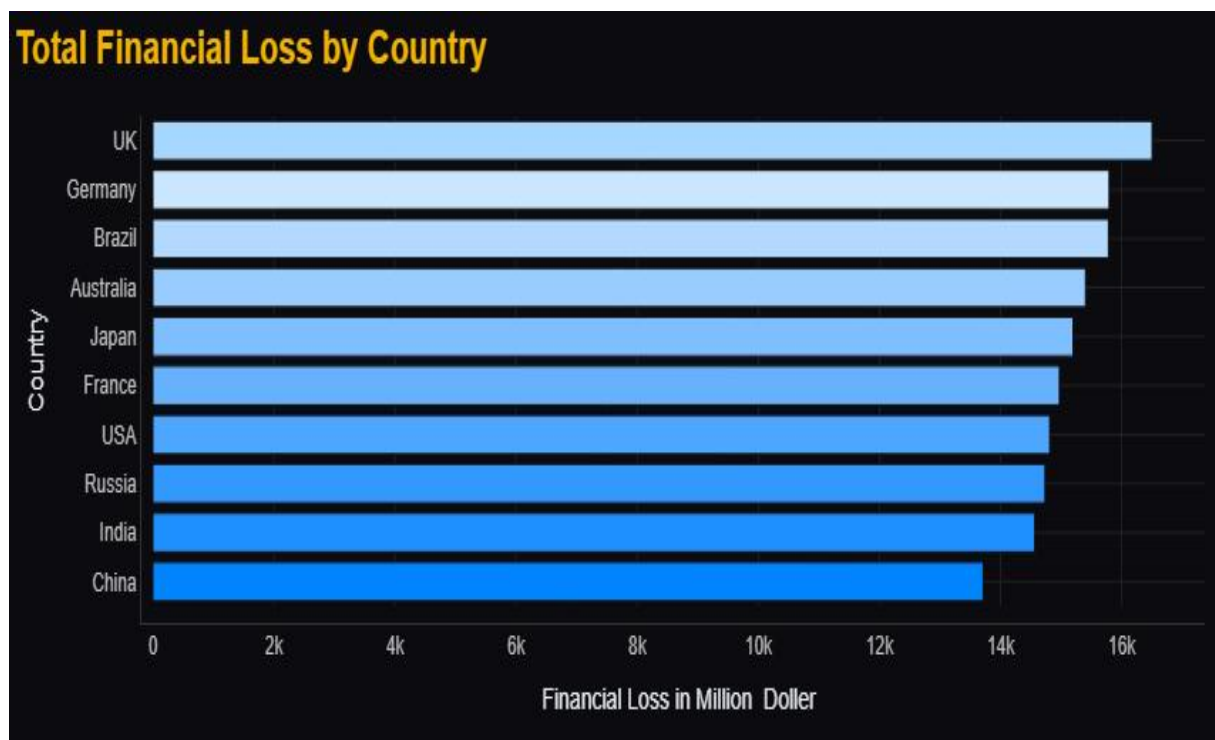
Overall, the KPI cards provide a clear overview of the dataset and serve as the starting point for detailed cyber crime analysis.

5.2 Bar Chart Analysis

The Bar Chart is used to compare different categories in the cyber crime dataset. It provides a clear visual representation of the number of cyber crime incidents across different countries, target industries, attack sources, or years. The height of each bar represents the corresponding value, making comparisons simple and effective.

This chart helps users quickly identify the categories with the highest and lowest number of cyber crime incidents. It also allows easy comparison between different groups, making trends and patterns more visible. Interactive features such as zooming and hovering provide additional details about each data point.

The Bar Chart improves the overall understanding of cyber crime trends and supports users in identifying high-risk areas that require stronger cyber security measures.



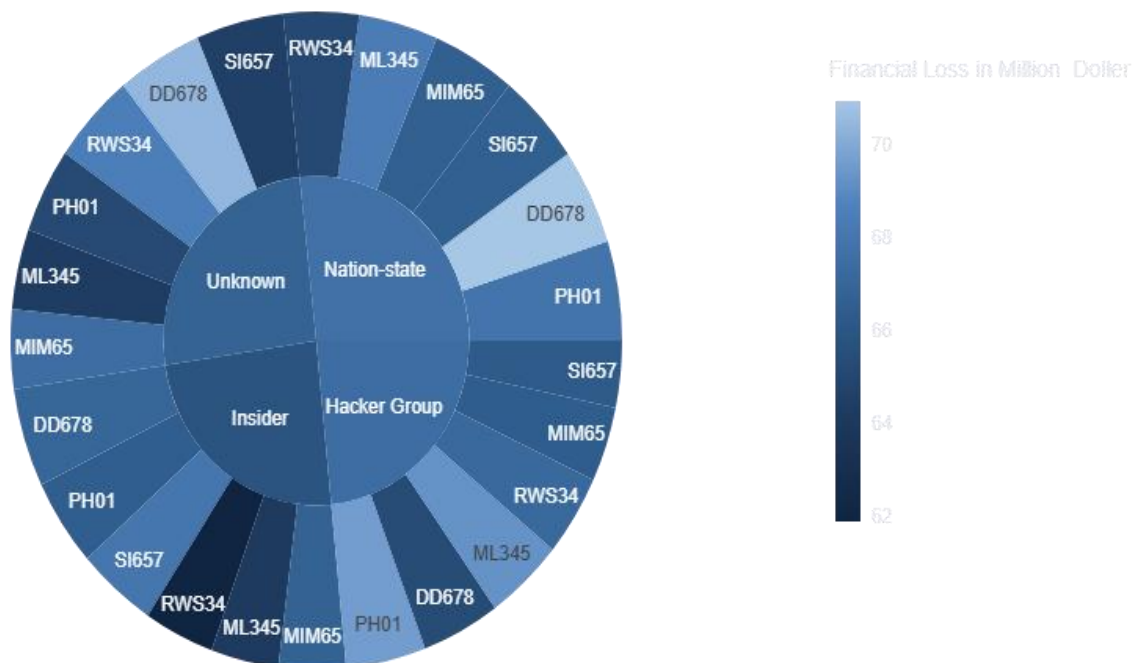
5.4 Sunburst Chart Analysis

The Sunburst Chart is used to display the hierarchical relationship between Attack Source and Attack ID based on the total financial loss. The inner circle represents the attack source, while the outer circle represents different attack IDs associated with each source.

This visualization helps users understand which attack sources contribute the highest financial losses and how different attack types are distributed within each source. The interactive nature of the chart allows users to click on a section and explore the data in greater detail.

The Sunburst Chart provides a clear hierarchical view of cyber attacks, making it easier to identify high-risk attack sources and their impact on financial losses.

Financial Loss by Attack Source & Attack Type



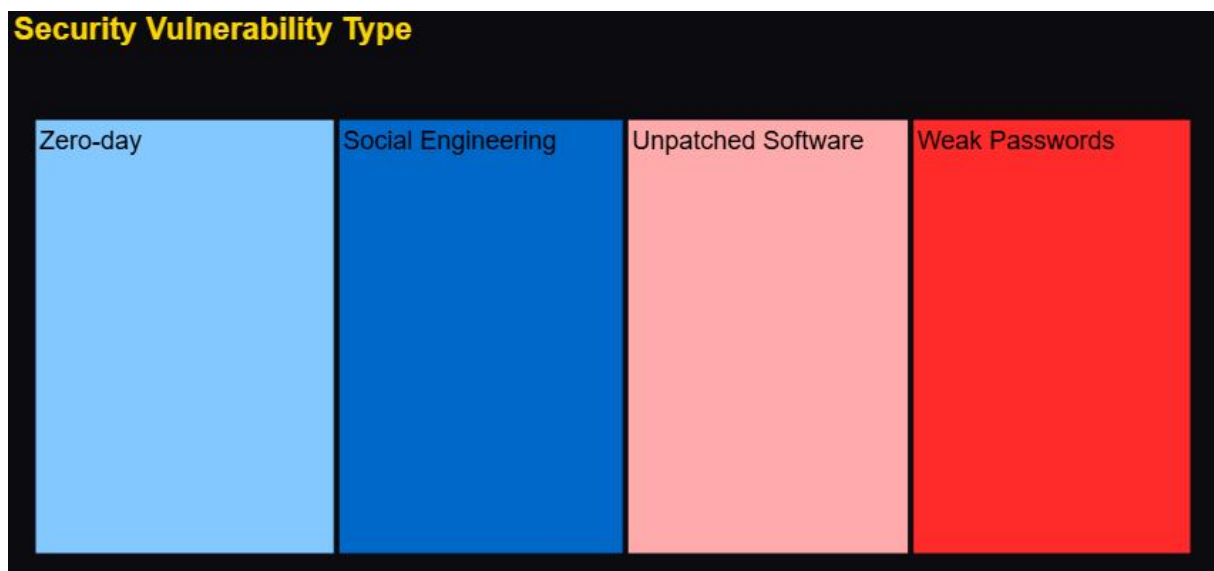
5.5 Treemap Analysis

The Treemap is used to visualize the distribution of Security Vulnerability Types in the cyber crime dataset. Each rectangle represents a different vulnerability type, and the size of the rectangle indicates its relative frequency or impact in the dataset.

This visualization helps users quickly identify the most common security vulnerabilities that contribute to cyber attacks. Larger rectangles represent vulnerabilities that occur more frequently, while smaller rectangles indicate less common vulnerabilities. The Treemap provides an efficient way to compare multiple categories within a limited space.

The interactive feature of the Treemap allows users to explore different vulnerability types by applying filters such as Country, Year, and Attack Source. This helps in understanding which security weaknesses require greater attention.

Overall, the Treemap simplifies complex cyber security data and supports better decision-making by highlighting the most significant security vulnerabilities.



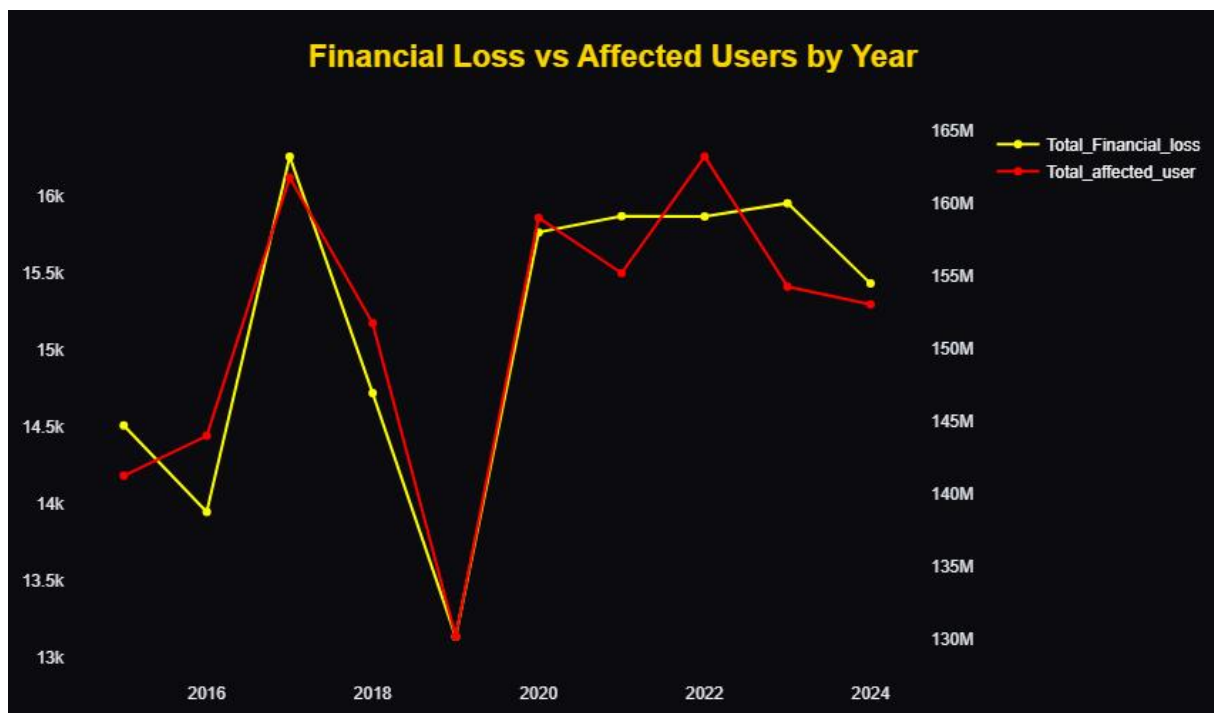
5.6 Financial Loss vs Affected Users by Year (Dual Line Chart Analysis)

The Dual Line Chart is used to compare Financial Loss and the Number of Affected Users over different years. The chart contains two separate lines, making it easy to observe changes and compare both parameters simultaneously.

This visualization helps identify whether an increase in cyber attacks also leads to higher financial losses and a greater number of affected users. It also highlights yearly trends, allowing users to understand how cyber crime has evolved over time.

The interactive features of the chart enable users to view detailed values for each year and analyze trends after applying different filters. This makes the comparison more effective and supports better interpretation of the data.

Overall, the Dual Line Chart provides valuable insights into the relationship between financial loss and affected users, helping organizations understand the overall impact of cyber crime and plan effective security measures.



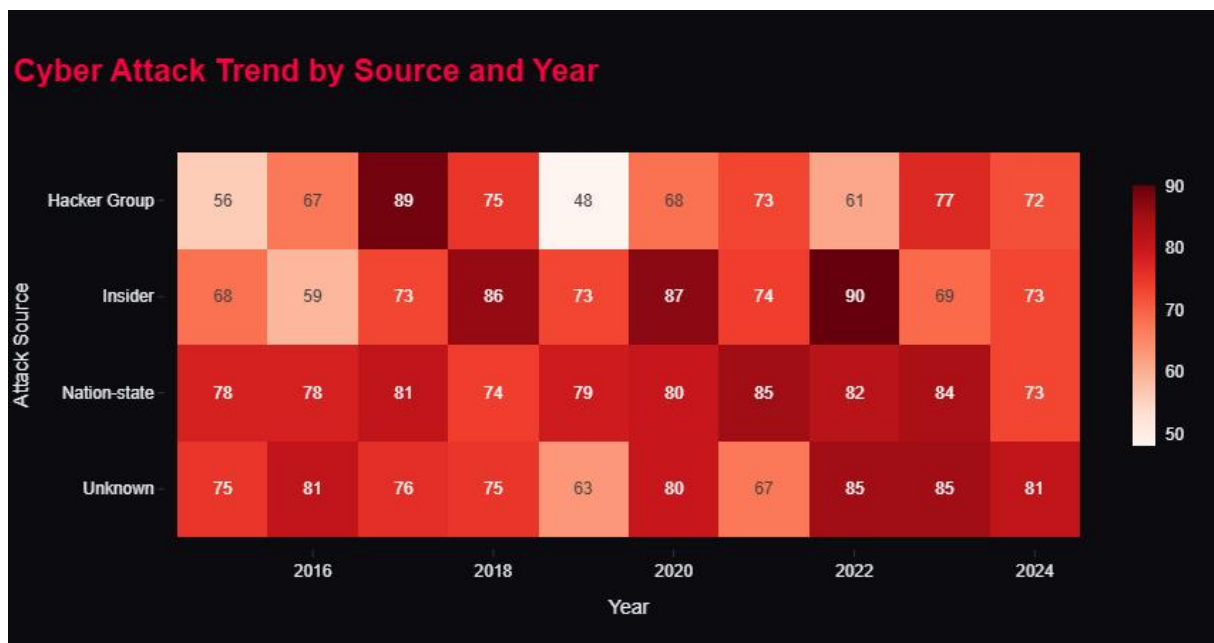
5.7 Heatmap Analysis

The Heatmap is used to analyze cyber attack trends based on Attack Source and Year. Different color intensities represent the number of cyber attacks, making it easy to identify periods and attack sources with higher incident rates.

The Heatmap provides a clear comparison of attack patterns across different years. Darker colors indicate a higher number of cyber attacks, while lighter colors represent fewer incidents. This visualization helps users quickly identify trends and changes in cyber attack activity over time.

The interactive Heatmap allows users to examine the relationship between attack sources and yearly cyber crime incidents. It supports better understanding of attack patterns and helps identify which attack sources are more active during specific years.

Overall, the Heatmap is an effective visualization for identifying cyber attack trends, comparing yearly data, and supporting data-driven cyber security decisions.



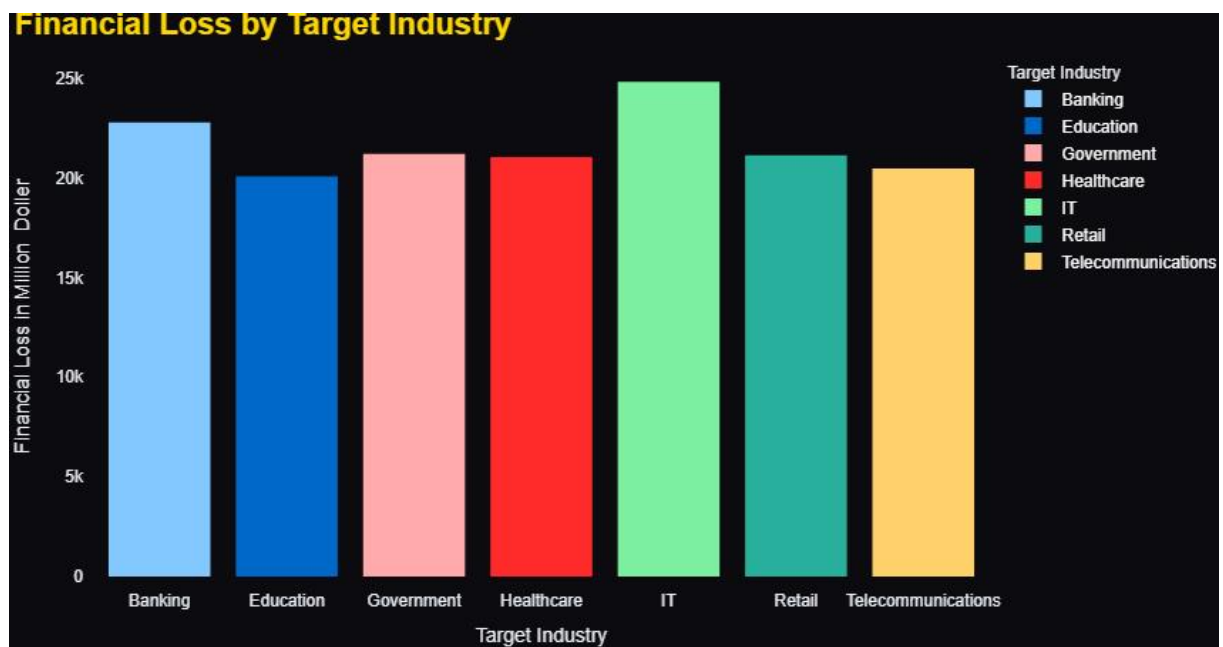
5.8 Financial Loss by Target Industry (Bar Chart Analysis)

The Bar Chart is used to compare the Financial Loss across different Target Industries affected by cyber attacks. Each bar represents an industry, and its height indicates the total financial loss incurred due to cyber crime incidents.

This chart helps users identify the industries that suffer the highest financial losses. By comparing different industries, organizations can recognize high-risk sectors and prioritize their cyber security strategies. The visualization also makes it easier to understand how cyber attacks impact various business sectors.

The interactive features of the chart allow users to apply filters such as Country, Year, and Attack Source, enabling detailed analysis for specific conditions. This improves the accuracy and usefulness of the analysis.

Overall, the Financial Loss by Target Industry Bar Chart provides valuable insights into the economic impact of cyber attacks on different industries and supports informed decision-making for improving cyber security.



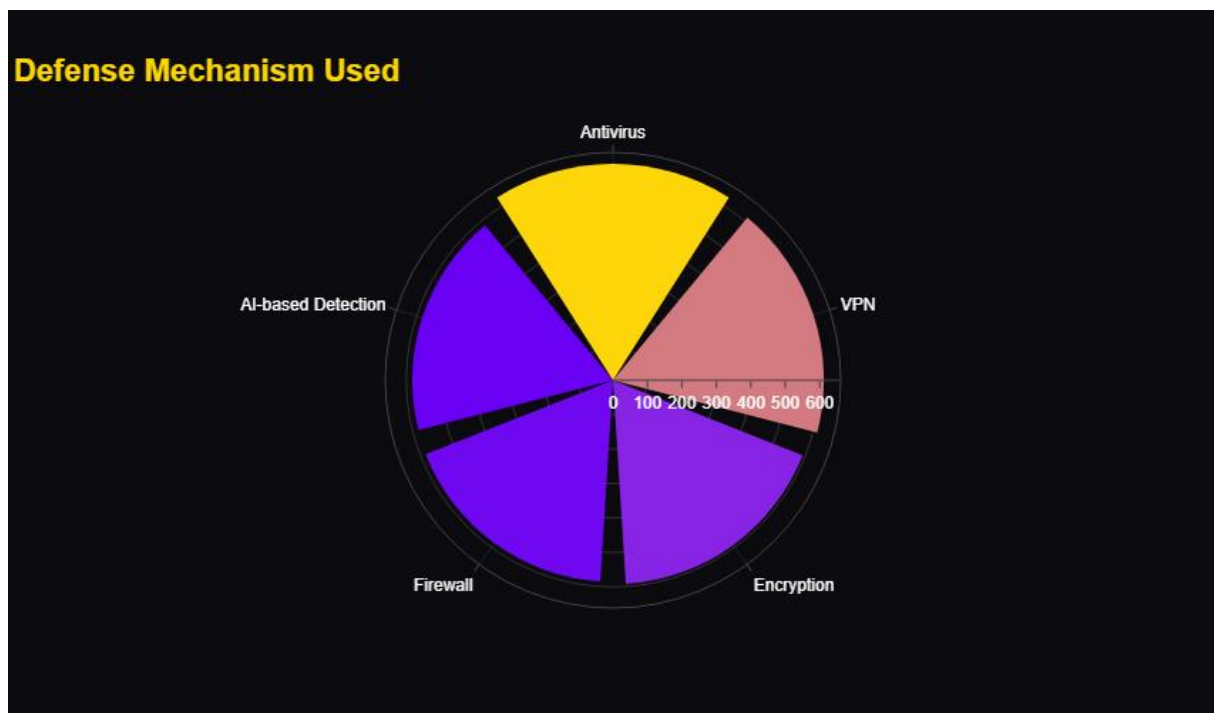
5.9 Defense Mechanism Used (Polar Bar Chart Analysis)

The Polar Bar Chart is used to visualize the distribution of different Defense Mechanisms used to protect systems from cyber attacks. Each bar represents a security mechanism, while its length indicates how frequently it is used in the dataset.

This visualization provides a clear comparison of various defense mechanisms such as Firewall, Antivirus, VPN, Encryption, and other security solutions. It helps users identify the most commonly used protection methods and understand their role in reducing cyber security risks.

The interactive nature of the Polar Bar Chart allows users to explore the data by applying filters such as Country, Year, and Target Industry. This makes it easier to compare defense strategies under different conditions and gain meaningful insights.

Overall, the Polar Bar Chart presents cyber security measures in an attractive and informative way, helping organizations evaluate their security practices and improve their defense against cyber threats.



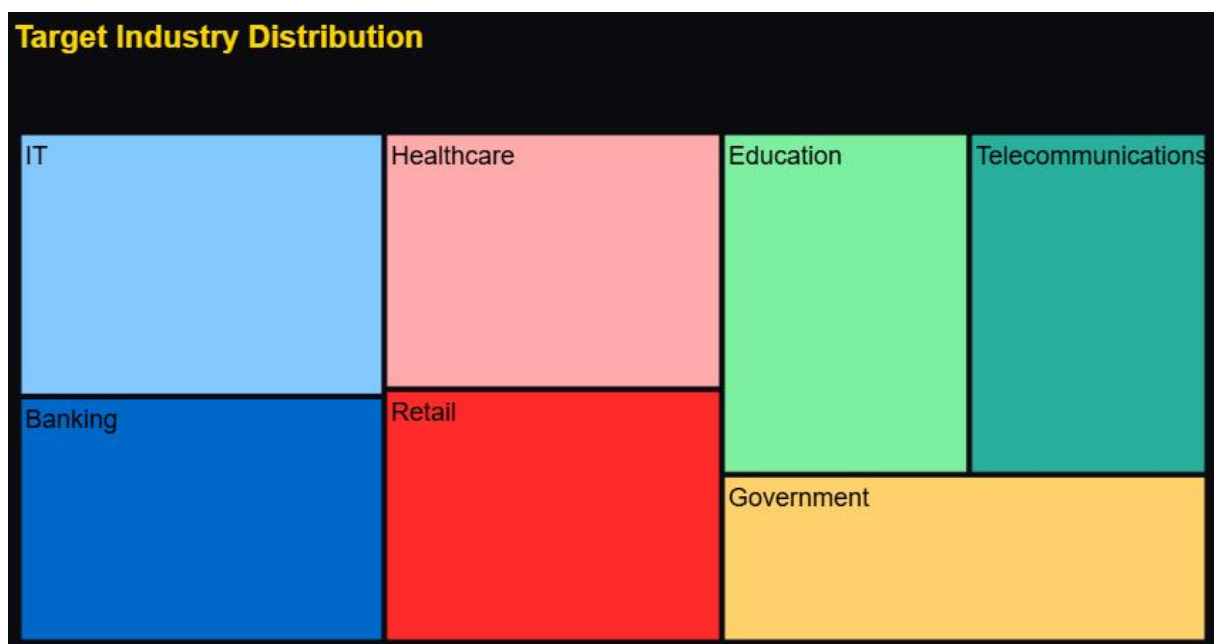
5.10 Target Industry Distribution (Treemap Analysis)

The Treemap is used to display the distribution of different Target Industries affected by cyber attacks. Each rectangle in the treemap represents a specific industry, and the size of the rectangle indicates its relative share in the dataset. This visualization makes it easy to compare multiple industries at a glance.

The Treemap helps users identify which industries are most frequently targeted by cyber criminals. Larger rectangles represent industries with a higher number of cyber attack incidents, while smaller rectangles indicate comparatively fewer attacks. This allows organizations to recognize high-risk sectors and strengthen their cyber security measures.

The interactive nature of the Treemap enables users to explore the data by applying filters such as Country, Year, and Attack Source. This provides a more detailed understanding of industry-wise cyber attack patterns and supports effective decision-making.

Overall, the Target Industry Distribution Treemap offers a clear and visually appealing representation of cyber attack distribution across different industries, helping users identify vulnerable sectors and improve cyber security planning.

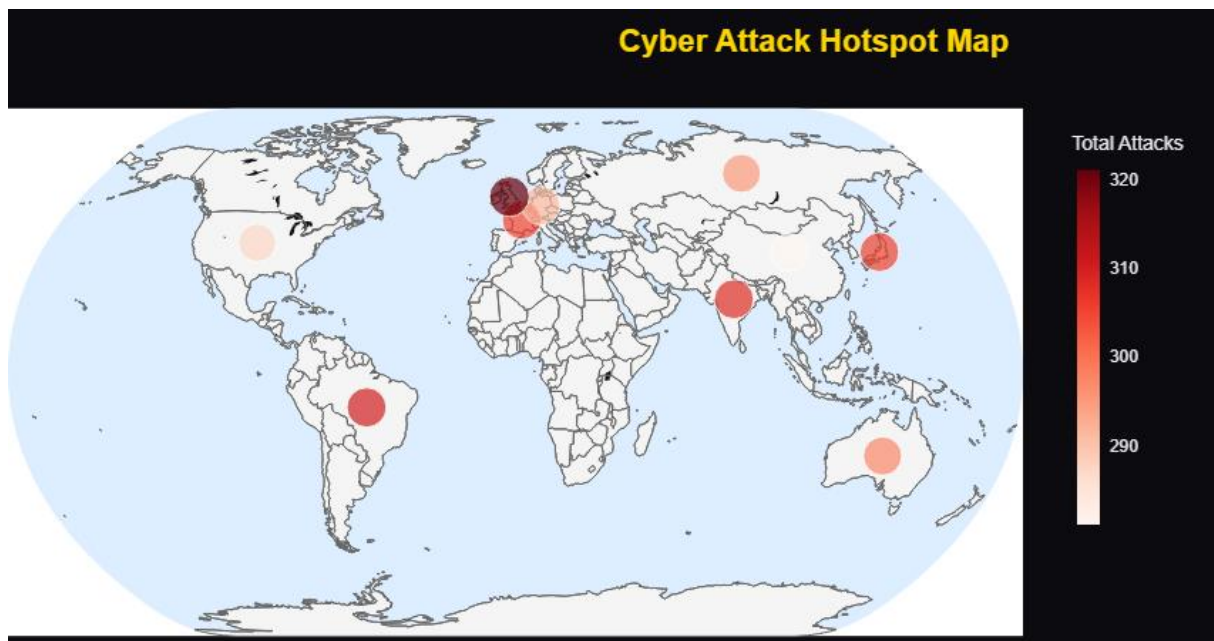


5.11 Cyber Crime Hotspot Map Analysis

The Cyber Crime Hotspot Map is used to visualize the geographical distribution of cyber attacks across different countries. The map highlights countries based on the total number of cyber attacks and financial losses. It helps users identify regions that are more affected by cyber crime.

The map provides an easy way to compare cyber crime incidents across different countries. Users can apply filters such as Country, Year, Target Industry, and Attack Source to analyze specific data. This visualization helps organizations understand regional cyber crime trends and supports better security planning.

Overall, the Cyber Crime Hotspot Map provides a clear geographical view of cyber attacks and helps users identify high-risk regions.



5.12 Insights Analysis

The Insights page summarizes the most important findings obtained from the cyber crime dataset. It displays information such as the country with the highest financial loss, the most targeted industry, the most common attack source, the most common security vulnerability, the most frequently used defense mechanism, and the total number of cyber attacks.

These insights provide a quick overview of the analysis without requiring users to examine every chart individually. They help users understand key cyber crime trends and support informed decision-making.

Overall, the Insights section makes the dashboard more informative by presenting the major findings in a simple and user-friendly format.

CHAPTER 6

CONCLUSION

The Cyber Crime Analysis Dashboard was successfully developed using Python, Pandas, Plotly, and Streamlit. The project analyzed cyber crime data and transformed it into meaningful visualizations that help users understand cyber attack patterns, financial losses, affected industries, and security vulnerabilities.

The dashboard provides interactive features such as filters, KPI cards, charts, treemaps, heatmaps, and a hotspot map, making data analysis simple and user-friendly. It enables users to explore cyber crime trends efficiently and supports better decision-making through clear visual insights.

Overall, this project demonstrates how data analytics and visualization techniques can simplify complex cyber crime data and improve cyber security awareness. It also highlights the importance of interactive dashboards in analyzing large datasets and presenting information in an effective manner.

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